# A Degree-Sum Theorem for Paths through Prescribed Vertices

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Joint work with Kun Cheng

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Connectivity $\kappa(G)$: the minimum number of vertices whose deletion disconnects $G$.

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**Theorem 1 (Chvátal–Erdős, 1972)**

$$\kappa(G) \geq \alpha(G) \implies G \text{ is hamiltonian;}$$

$$\kappa(G) > \alpha(G) \implies G \text{ is hamiltonian-connected.}$$

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**Theorem 2 (Ota, 1995)**

Let $G$ be a $k$-connected graph of order $n$, where $k \geq 2$. If

$$\sigma_{p+1}(G) \geq n + p(p-1)$$

for every $k \leq p < \alpha(G)$, then $G$ is a Hamiltonian graph.

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**Theorem 3 (Ota, 1995)**

Let $G$ be a graph of order $n$, and let $S \subseteq V(G)$. Suppose $\kappa(S) \geq k \geq 2$. If

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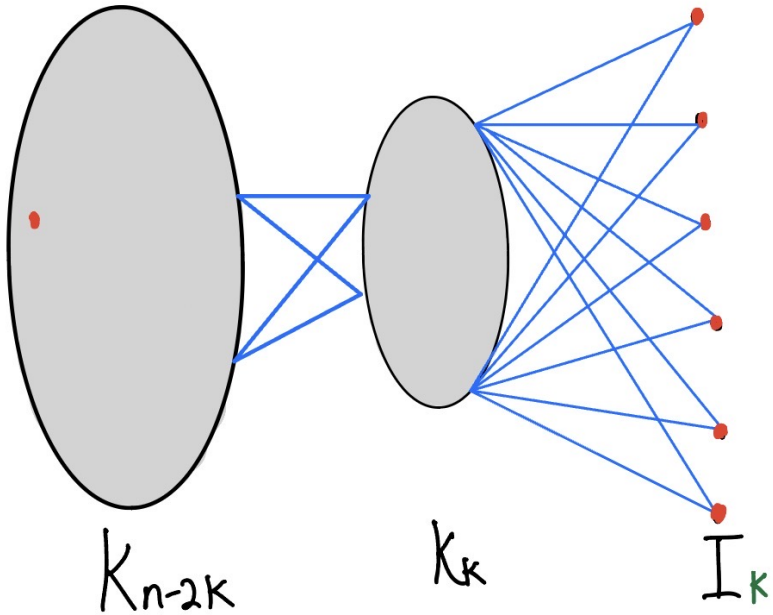
for every integer $k \leq p < \alpha(S)$, then $G$ contains a cycle through all vertices of $S$.

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non-hamiltonian graph

The degree sum of the red vertices is:

$n-k-1+k^2$

**Theorem 4 ( Zhan and Z., 2022)**

Let $f(n,k)$ denote the maximum size of a nonhamiltonian graph of order $n$ and connectivity $k$. Then

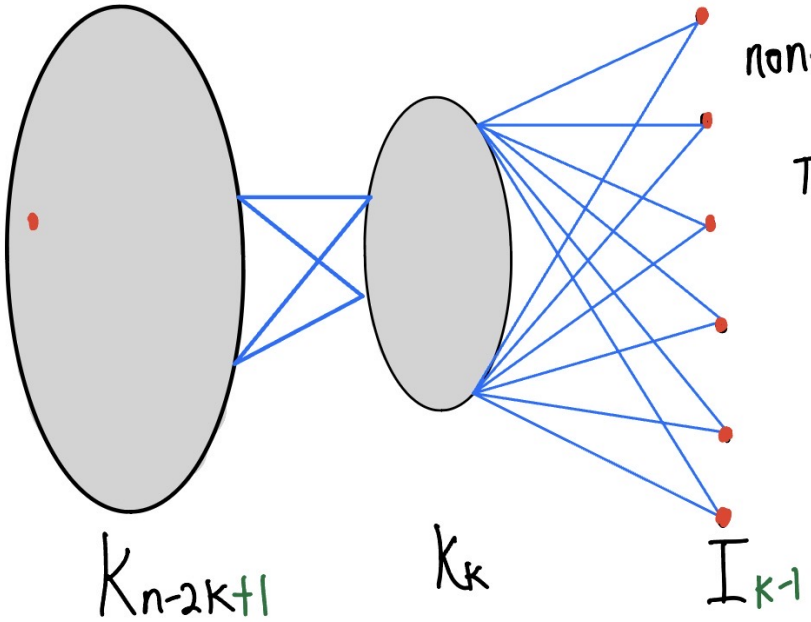
$$f(n,k)=\begin{cases}\binom{n-k}{2}+k^2 & \text{if } n \text{ is odd and } n\geq 6k-5 \text{ or}\\ & n \text{ is even and } n\geq 6k-8,\\ \frac{3n^2-8n+5}{8}+k & \text{if } n \text{ is odd and } 2k+1\leq n\leq 6k-7,\\ \frac{3n^2-10n+16}{8}+k & \text{if } n \text{ is even and } 2k+2\leq n\leq 6k-10.\end{cases}$$

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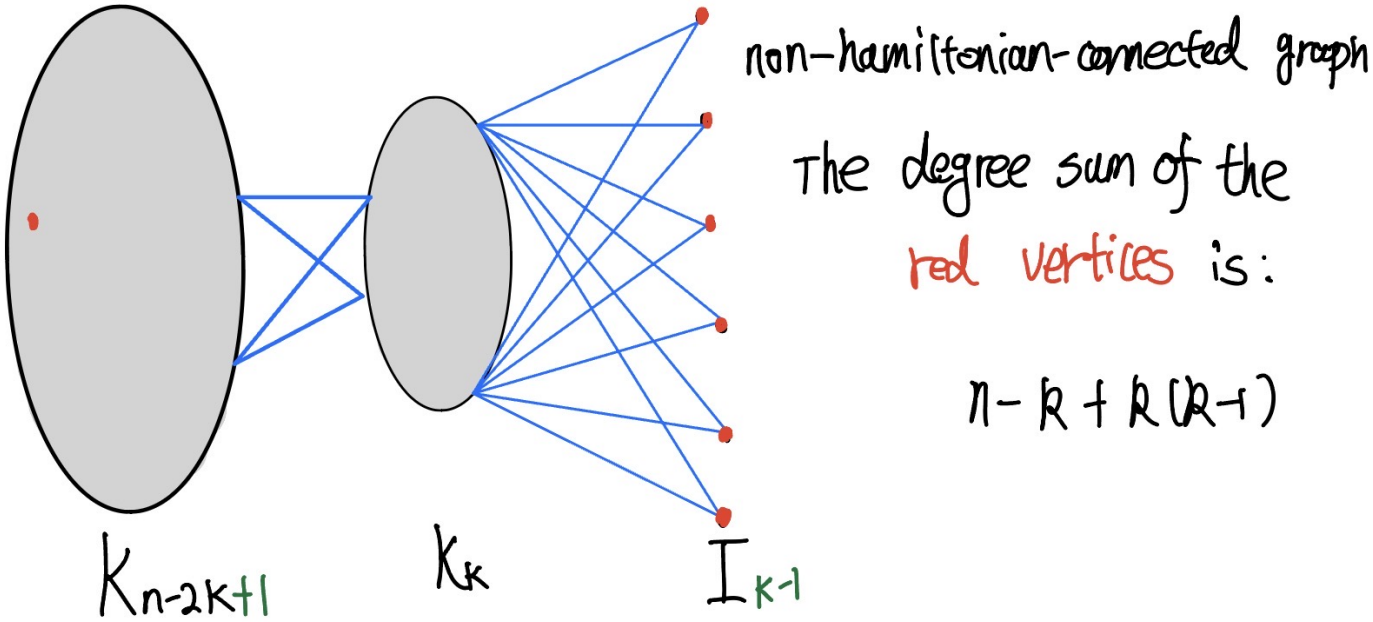
What is the corresponding result for Hamiltonian-connectedness?




non-hamiltonian-connected graph

The degree sum of the red vertices is:

$$n-k+k(k-1)$$




**Theorem 5 (Cheng and Z., 2026$^+$)**

Let $G$ be a $k$-connected graph of order $n$, where $k \geq 2$. If

$$\sigma_p(G) \geq n - p + p(p-1) + 1 \text{ for every integer } k \leq p \leq \alpha(G),$$

then $G$ is hamiltonian-connected.

## Our theorem: two prescribed endpoints

**Theorem 6 (Cheng and Z., $2026^+$)**

Let $G$ have order $n$, and let $S \subseteq V(G)$, with $|S| \geq 2$. Suppose $\kappa(S) \geq k \geq 2$ and

$$\sigma_p(S) \geq n + (p-1)^2 \text{ for every } k \leq p \leq \alpha(S).$$

Then every two distinct $u, v \in S$ are the endpoints of a path containing all vertices of $S$.

The path may use vertices outside $S$.

## The cycle and path conditions

### Ota's Theorem

$$\sigma_p(S) \geq n + (p-1)(p-2), \qquad \alpha(S) \geq p \geq k+1.$$

A cycle containing $S$.

### Our Theorem

$$\sigma_p(S) \geq n + (p-1)^2, \qquad \alpha(S) \geq p \geq k.$$

A path containing $S$ with any prescribed endpoints in $S$.

**Theorem 7 (Cheng and Z., 2026$^+$)**

Let $f(n,k)$ denote the maximum number of edges in a graph of order $n$ and connectivity $k$ that is not hamiltonian-connected, where $k \geq 2$ and $n \geq 2k$. Then

$$f(n,k) = \begin{cases} \binom{n-k+1}{2} + k(k-1) & \text{if } n \text{ odd and } n \geq 6k-13 \text{ or} \\ & \quad n \text{ even and } n \geq 6k-10, \\ \frac{3n^2-6n}{8} + k & \text{if } n \text{ even and } 2k \leq n \leq 6k-12, \\ \frac{3n^2-8n+13}{8} + k & \text{if } n \text{ odd and } 2k+1 \leq n \leq 6k-15. \end{cases}$$

**Conjecture 9 (Faudree-Faudree-Gould-Jacobson-Magnant, 2010)**

Let $G$ be a $k$-connected graph of order $n$, where $k \geq 3$. If

$$\delta(G) > \frac{n + k^2 - 2k}{k}, \qquad \delta(G) \geq \alpha(G) + k - 2,$$

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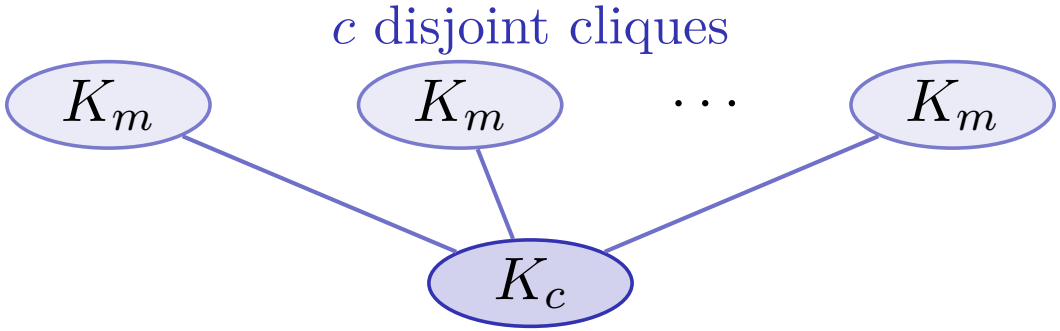
then $G$ is hamiltonian-connected.

Previously known for sufficiently large $n$:

- $k = 3, 4$ with $\kappa(G) \geq k$;
- general $k$ with $\kappa(G) \geq 4k^2 + 1$.

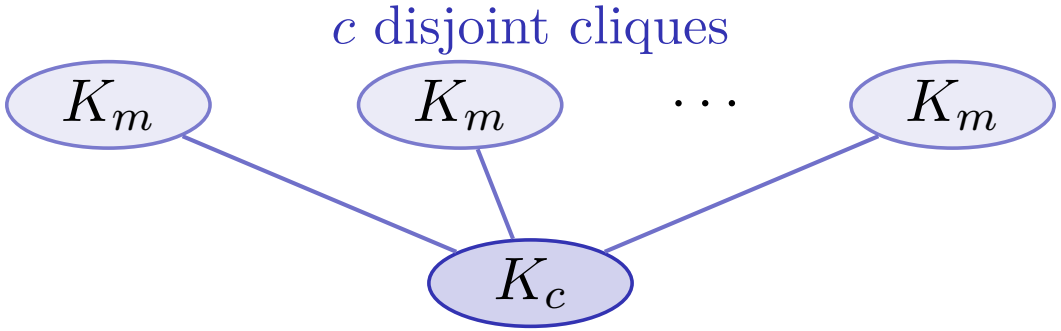
# Sharpness of the conjecture

Let $H(c, m) = K_c \vee cK_m$.



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| **Condition** | **Counterexample** | |
|---|---|---|
| $\delta(G) > \dfrac{n+k^2-2k}{k}$ | $H(k,m), \quad m \geq k-1$ | $\delta(G) = \dfrac{n+k^2-2k}{k}$ |
| $\kappa(G) \geq k$ | $H(k-1,m), \quad m \geq k$ | $\kappa(G) = k-1$ |
| $\delta(G) \geq \alpha(G)+k-2$ | $H(t,k-2), \quad t \geq k+1$ | $\delta(G) = \alpha(G)+k-3$ |

Each example satisfies the other two conditions and is not hamiltonian-connected.

## Proof of Conjecture 9

For every $k \leq p \leq \alpha(G)$, the assumptions give

$$k\delta(G) \geq n + (k-1)^2, \qquad \delta(G) \geq p + k - 2.$$

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Hence

$$\begin{aligned}\sigma_p(G) &\geq p\delta(G) = k\delta(G) + (p-k)\delta(G) \\ &\geq n + (k-1)^2 + (p-k)(p+k-2) \\ &= n + (p-1)^2.\end{aligned}$$

Our theorem with $S = V(G)$ gives Hamiltonian-connectedness.

1. V. Chvátal and P. Erdős, A note on Hamiltonian circuits, *Discrete Math.* **2** (1972), 111–113.

2. J. R. Faudree, R. J. Faudree, R. J. Gould, M. S. Jacobson and C. Magnant, Chvátal–Erdős type theorems, *Discuss. Math. Graph Theory* **30** (2010), 245–256.

3. K. Ota, Cycles through prescribed vertices with large degree sum, *Discrete Math.* **145** (1995), 201–210.

4. X. Zhan and L. Zhang, The maximum size of a nonhamiltonian graph with given order and connectivity, *Discrete Math.* **346** (2023), 113208.

*Thank you!*